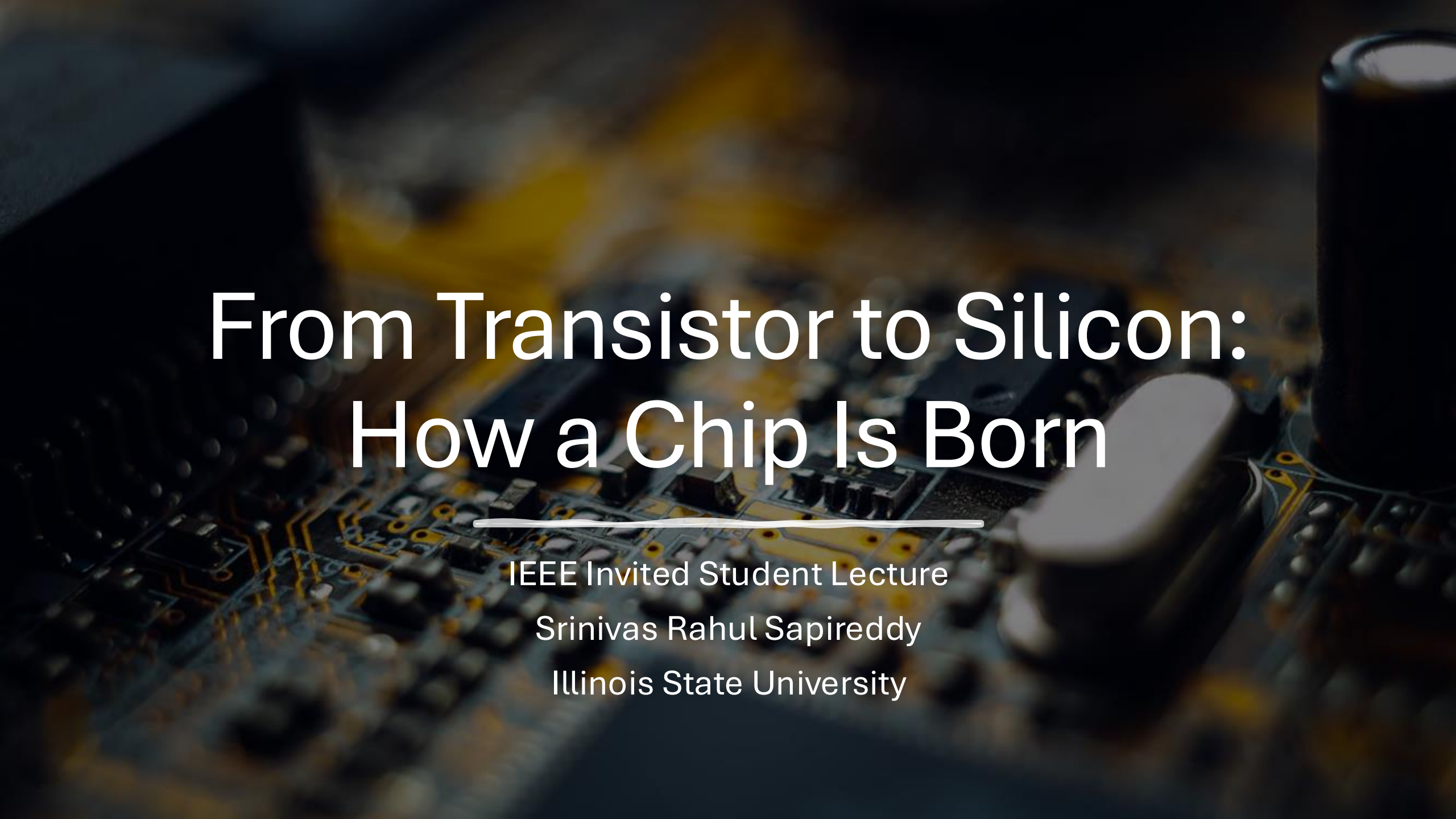




From Transistor to Silicon: How a Chip Is Born

IEEE Invited Student Lecture

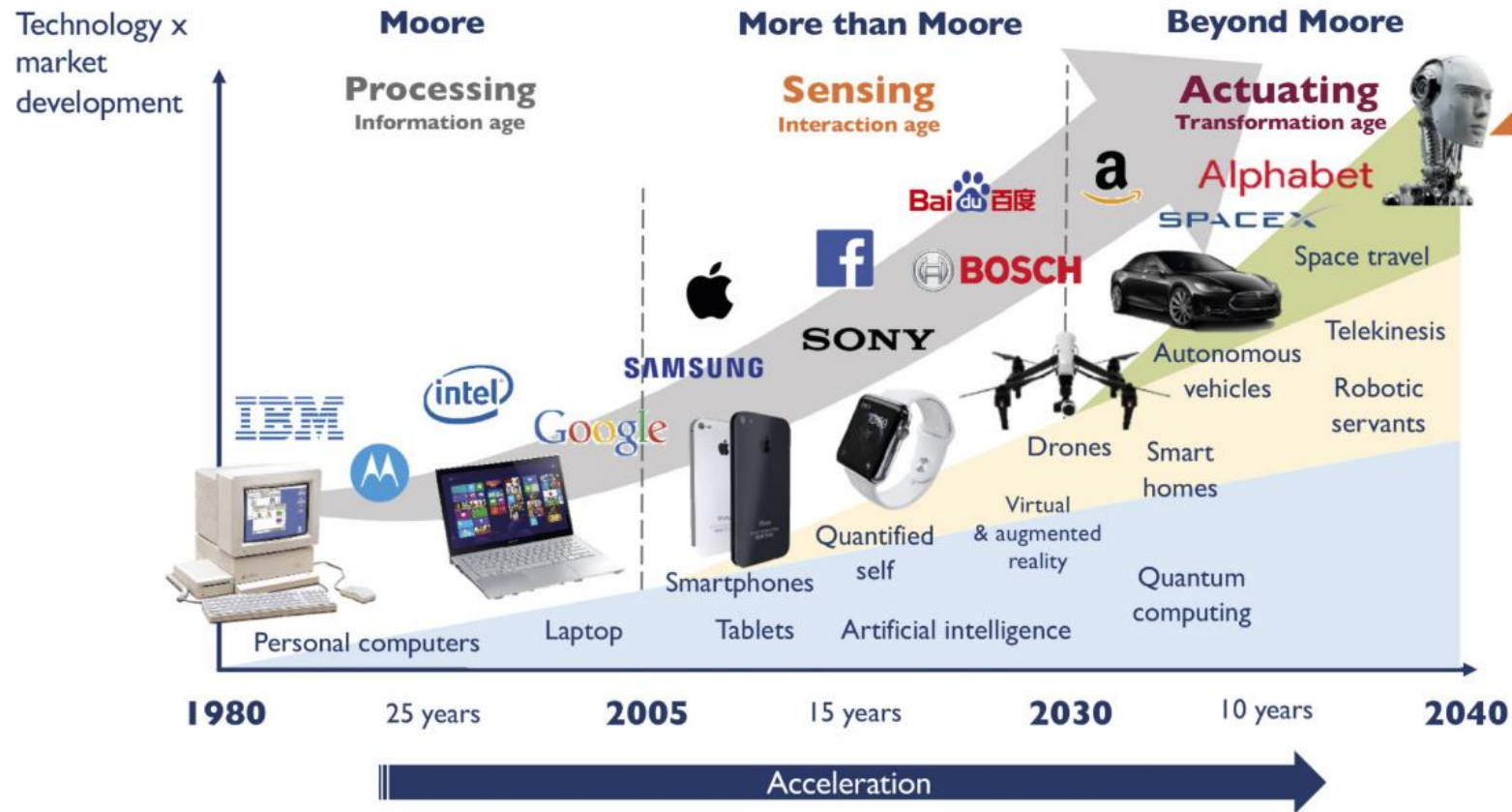
Srinivas Rahul Sapireddy

Illinois State University

Why Chips Matter

The Invisible Technology Around Us

Every smart device runs on chips.



From PCs to AI robots, every revolution is powered by silicon chips.

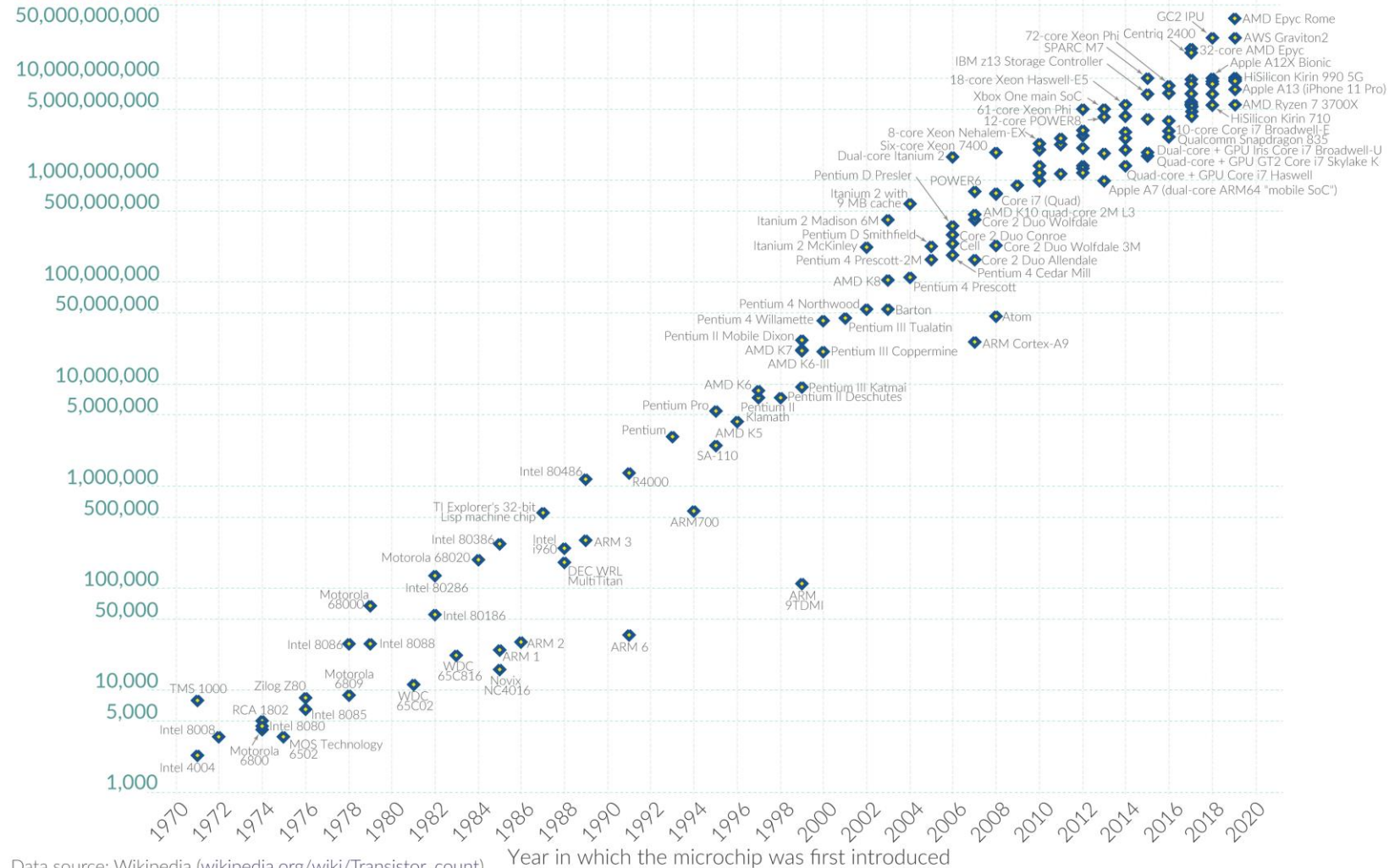
Moore's Law

Moore's Law: The number of transistors on microchips doubles every two years

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years. This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.

Our World
in Data

Transistor count



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)

OurWorldinData.org – Research and data to make progress against the world's largest problems.

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The Smallest Building Block

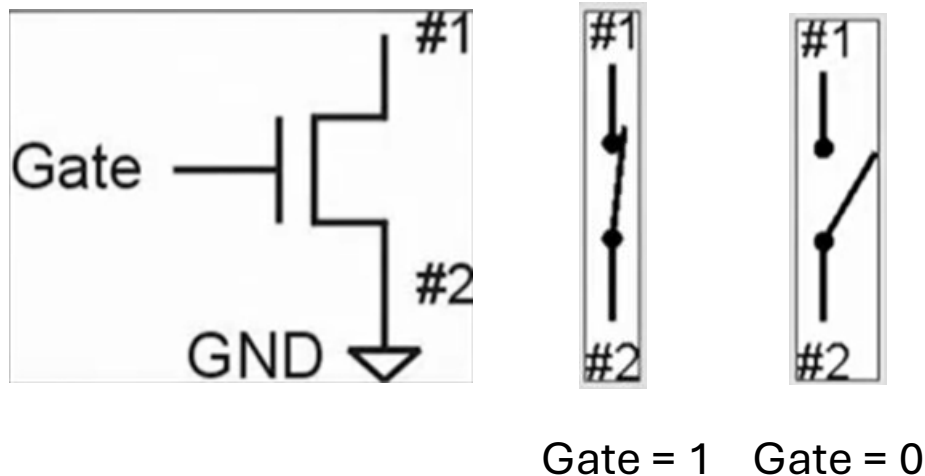
A transistor is an electronic switch.

n-type MOS Transistor

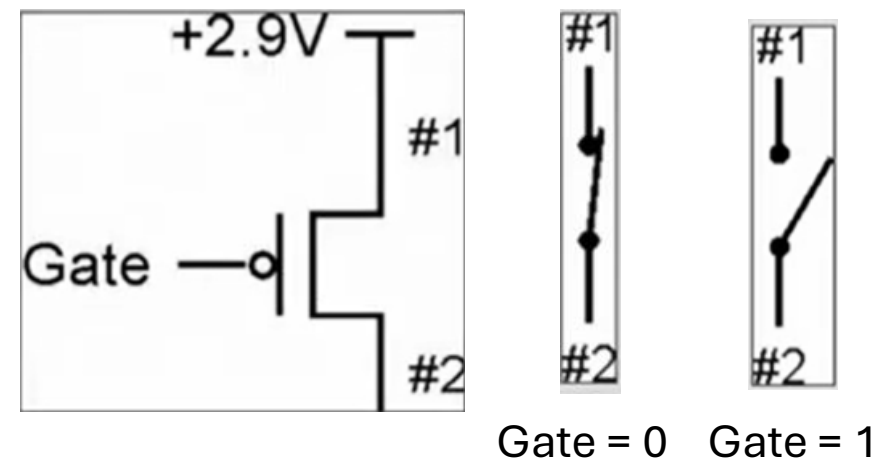
- If the **Gate = 1 (positive voltage)**, current can flow from **source to drain** → **switch ON (closed)**
- If the **Gate = 0 (zero voltage)**, current cannot flow → **switch OFF (open)**

p-type MOS Transistor

- **Gate = 0** → **ON** → switch closed
- **Gate = 1** → **OFF** → switch open



n-type MOS Transistor



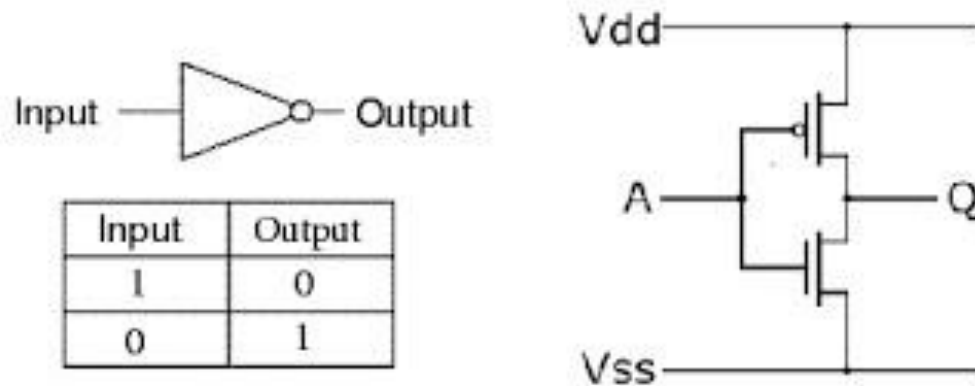
p-type MOS Transistor

Basic Logic Gates

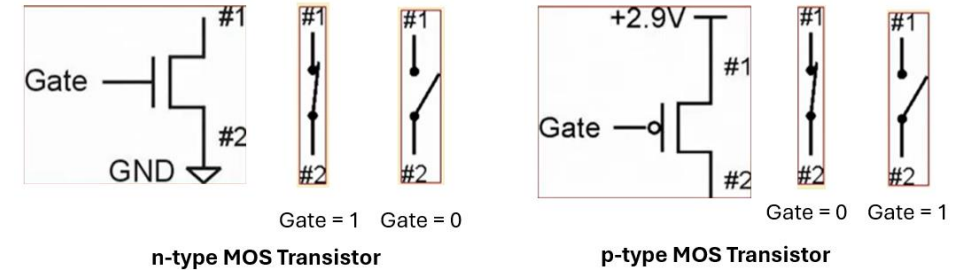
- Digital logic gates typically operate on **binary logic**.
- Two distinct values in binary logic, denoted by **0 and 1**.

Why binary logic?

- It is easy to design electronic circuits with **two distinct states**.
- Examples:
 - An electronic switch is either **open or closed**.



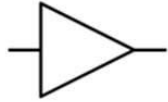
How Transistors Build an NOT Gate (Inverter)



- Input A = 0 $\rightarrow$ Output Q = 1
- Input A = 1 $\rightarrow$ Output Q = 0
- PMOS on top, NMOS on bottom
- One transistor ON, the other OFF

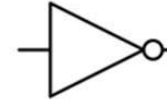
Basic Logic Gates

Buffer



Input	Output
0	0
1	1

Inverter



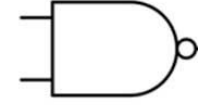
Input	Output
0	1
1	0

AND



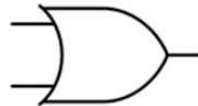
A	B	Output
0	0	0
1	0	0
0	1	0
1	1	1

NAND



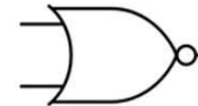
A	B	Output
0	0	1
1	0	1
0	1	1
1	1	0

OR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	1

NOR



A	B	Output
0	0	1
1	0	0
0	1	0
1	1	0

XOR



A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

XNOR

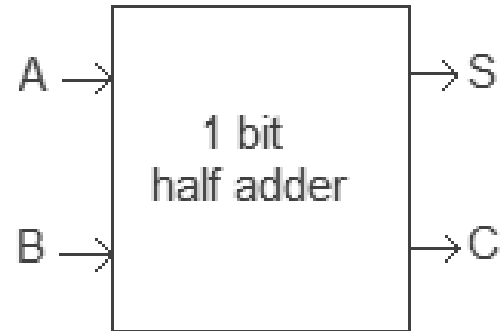


A	B	Output
0	0	1
1	0	0
0	1	0
1	1	1

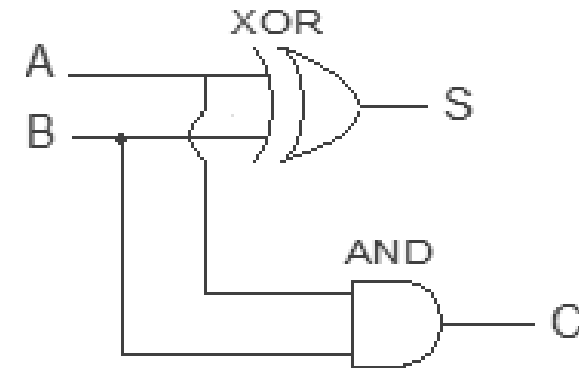
From Gates to Circuits

Inputs		Outputs	
A	B	S	C
0	0	0	0
1	0	1	0
0	1	1	0
1	1	0	1

Truth table



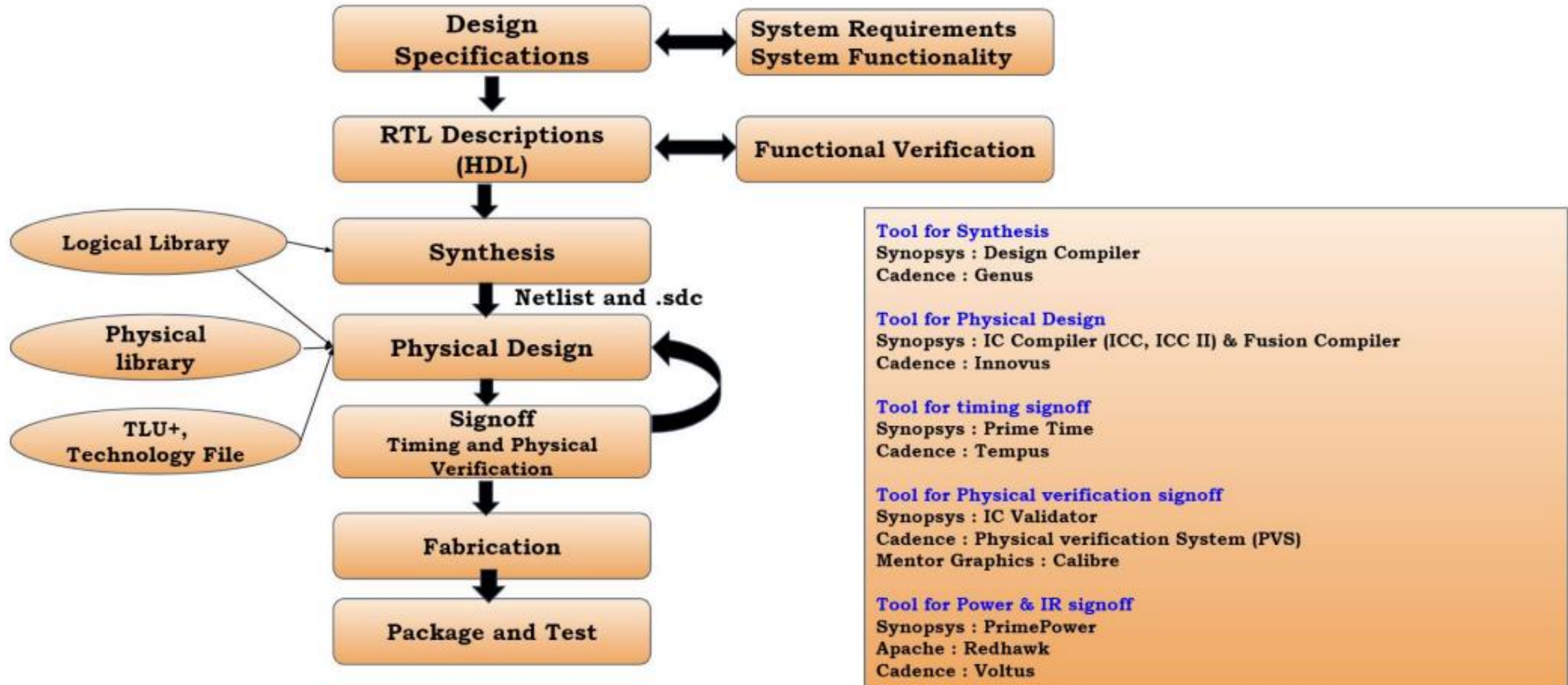
Schematic



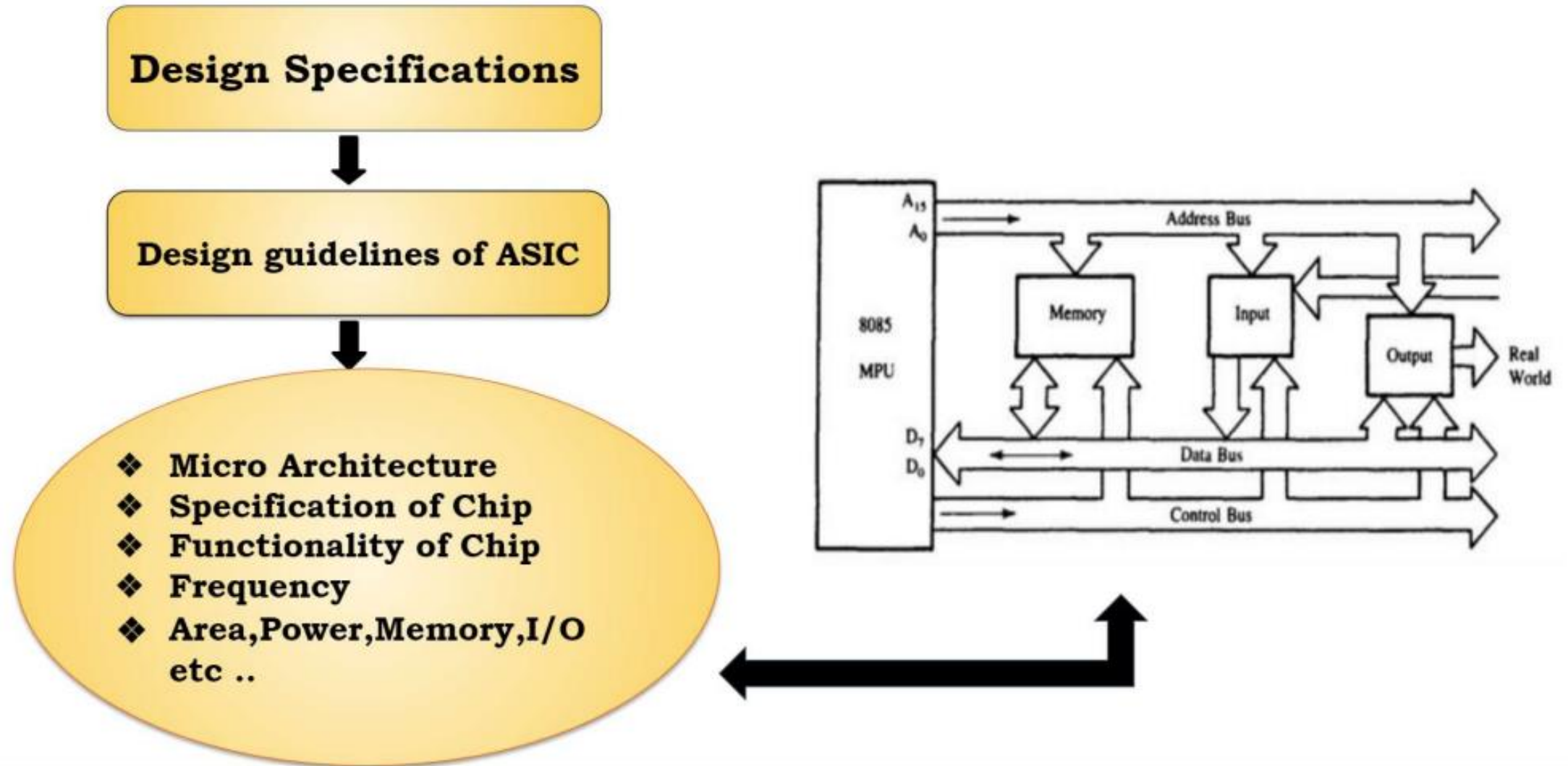
Realization

Half Adder Circuit Realization

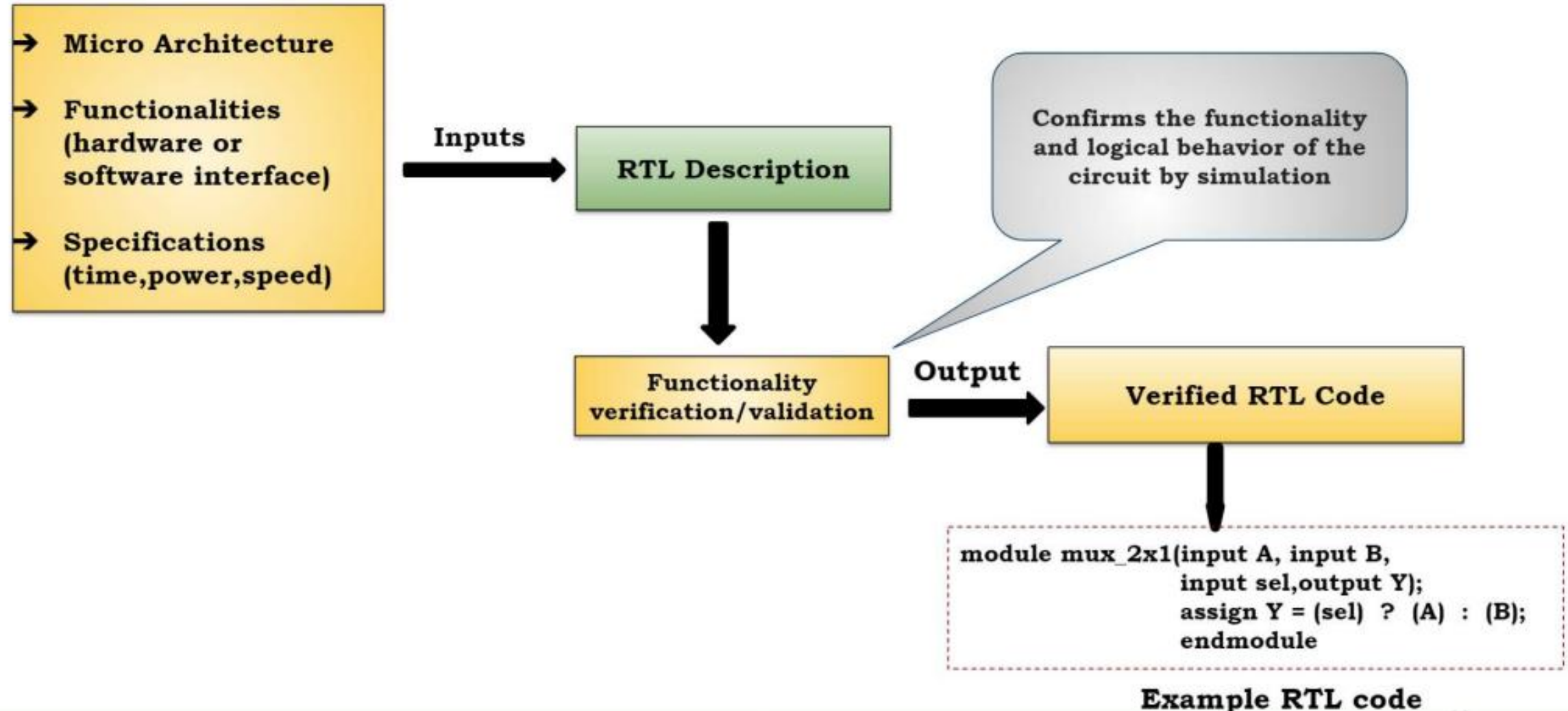
ASIC Design Flow



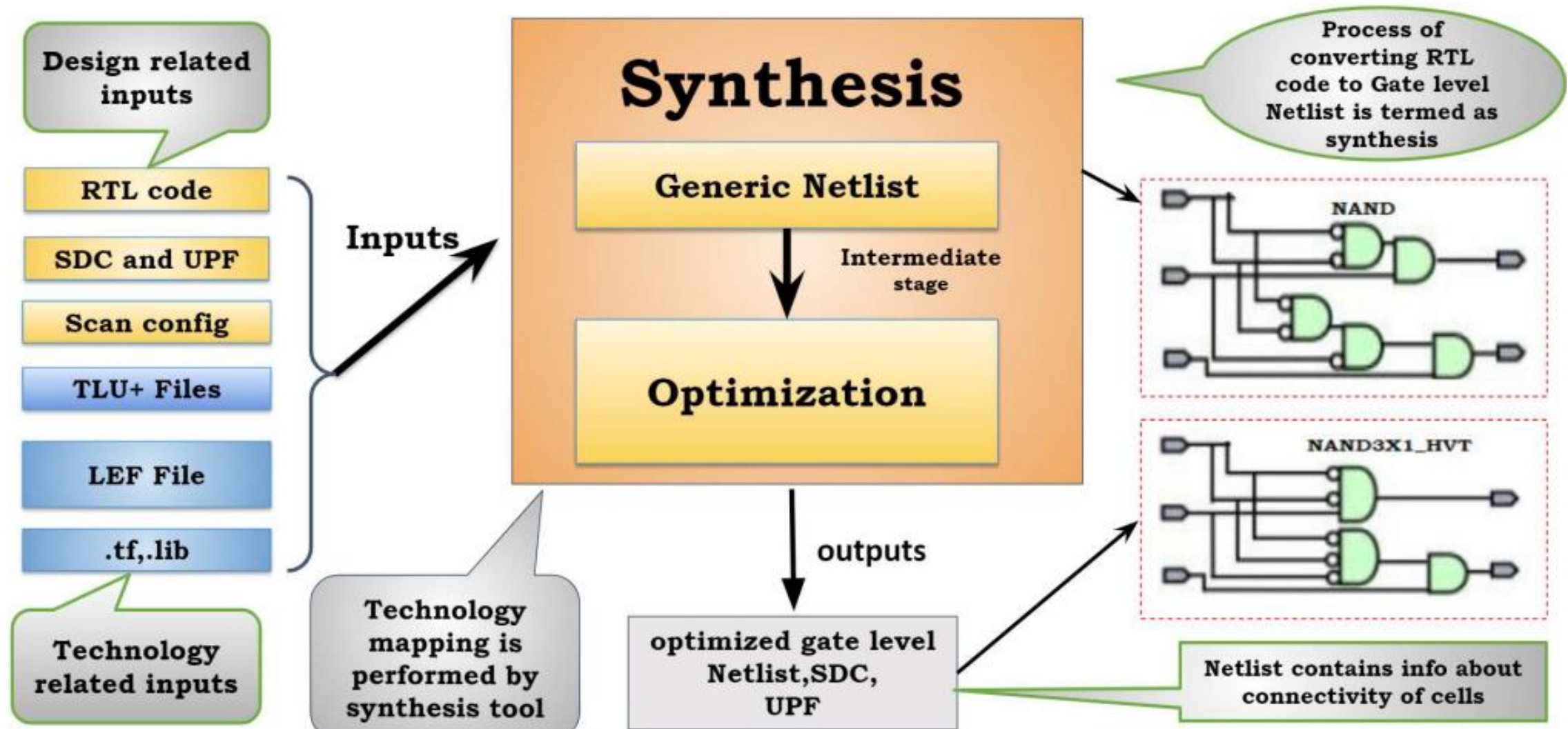
Design Specifications



RTL Description (HDL)



Synthesis



RTL Code to Optimized Netlist

RTL Code

```
module Mux2_1 (In, sel, out);  
  
input [1:0] In;  
input sel;  
  
output out;  
  
assign out = sel ? In[1] : In[0];  
  
endmodule
```

RTL code
converted to
Intermediate
or Generic
Netlist

Generic Netlist

```
module Mux2_1 ( In, sel, out );  
input [1:0] In;  
input sel;  
output out;  
wire  N0, N1, N2;  
  
SELECT_OP C11 ( .DATA1(In[1]), .DATA2(In[0]),  
                .Z(out) ); .CONTROL1(N0), .CONTROL2(N1),  
  
GTECH_BUF B_0 ( .A(sel), .Z(N0) );  
GTECH_BUF B_1 ( .A(N2), .Z(N1) );  
GTECH_NOT I_0 ( .A(sel), .Z(N2) );  
endmodule
```

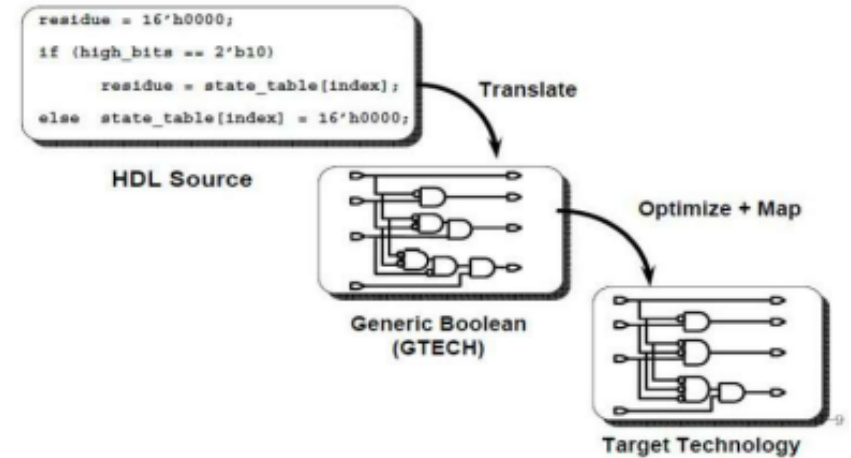
Final Netlist

```
module Mux2_1 ( In, sel, out );  
input [1:0] In;  
input sel;  
output out;  
  
MUX21X1_HVT U3 ( .A1(In[0]),  
                .A2(In[1]), .S0(sel), .Y(out) );  
endmodule
```

Generic to
Optimized
Gate level
Netlist

Logic Synthesis

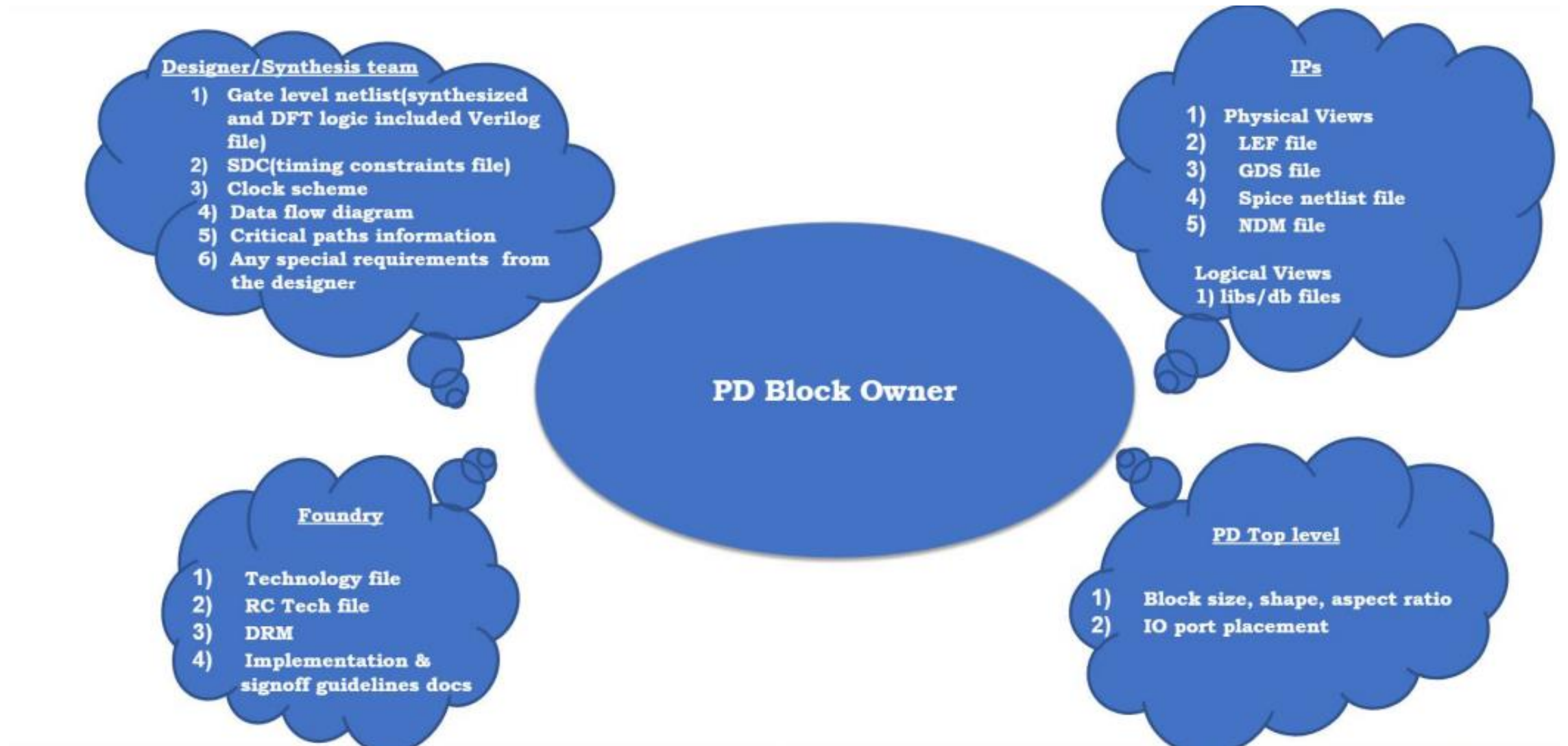
- **Synthesis** is an automatic method of converting a higher level of abstraction to a lower level of abstraction.
- In other words, the synthesis process converts Register Transfer Level (RTL) description to gate-level net-list.
- The gate-level netlist can be optimized for area, speed, testability, etc.
- Synthesis goals : To generate a gate level netlist to meet
 - Performance(Timing)
 - Power
 - Area
 - Routability



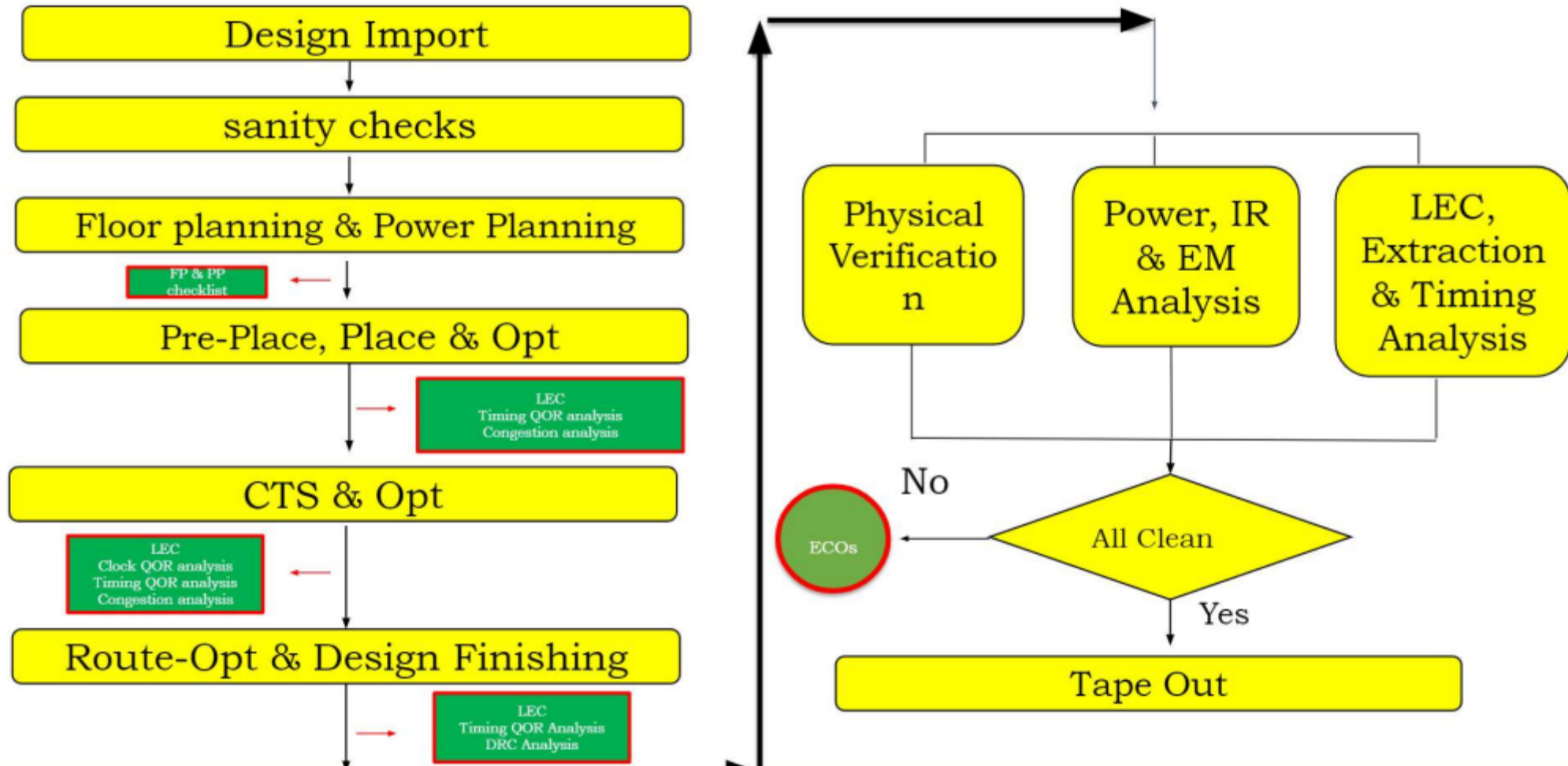
RTL	Generic Netlist	Synthesized Netlist
<code>assign z = x y;</code>	<code>or I2 (z, x, y);</code>	<code>OR2X1 I2 (.Y(z), .A(x), .B(y));</code>
<code>assign z = x & y;</code>	<code>and I3 (z, x, y);</code>	<code>AND2X2 I3 (.Y(z), .A(x), .B(y));</code>

PD FLOW and Terminology

Inputs to PD Block Implementation



ASIC Block-level PD Flow

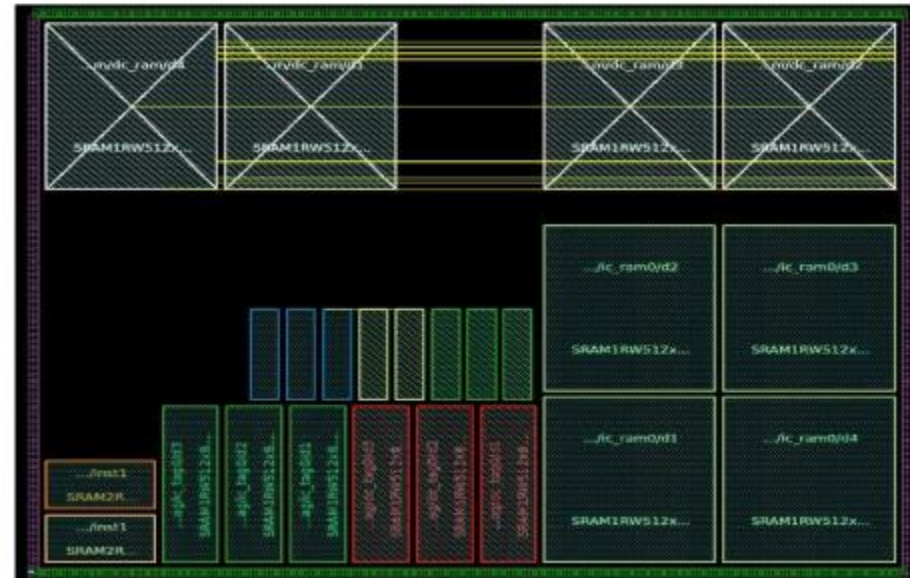


Full Chip Partitioning

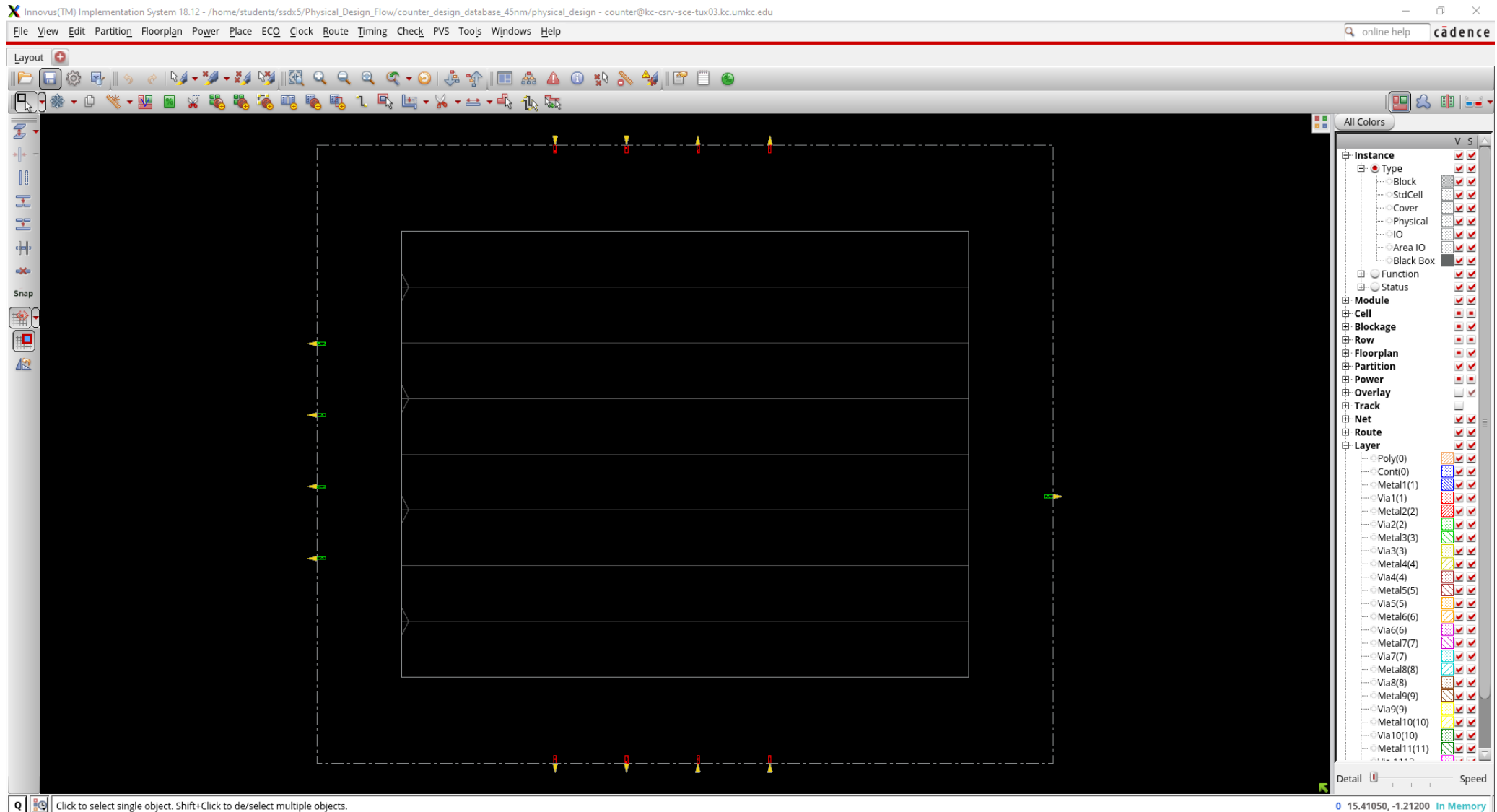
- Partitioning divides a large circuit into group of smaller blocks.
- These blocks can be designed independently and simultaneously to speed up the design process.
- An ASIC design can be partitioned logically or Physically.
 - Logical partitioning is done in the early stages, the design is partitioned according to its logical functions.
 - Physical partitioning is performed during the physical design activity.
- The partitioned blocks are implemented independently at block level and the outputs are provided for integration at full-chip level

Floorplan

- Plan the size, shape of the block
- Plan the positions of IO ports.
- Plan the positions of the modules/macros at the beginning of the design cycle to optimize the circuit performance in terms of chip area, wire length, routability and others.



Floorplan



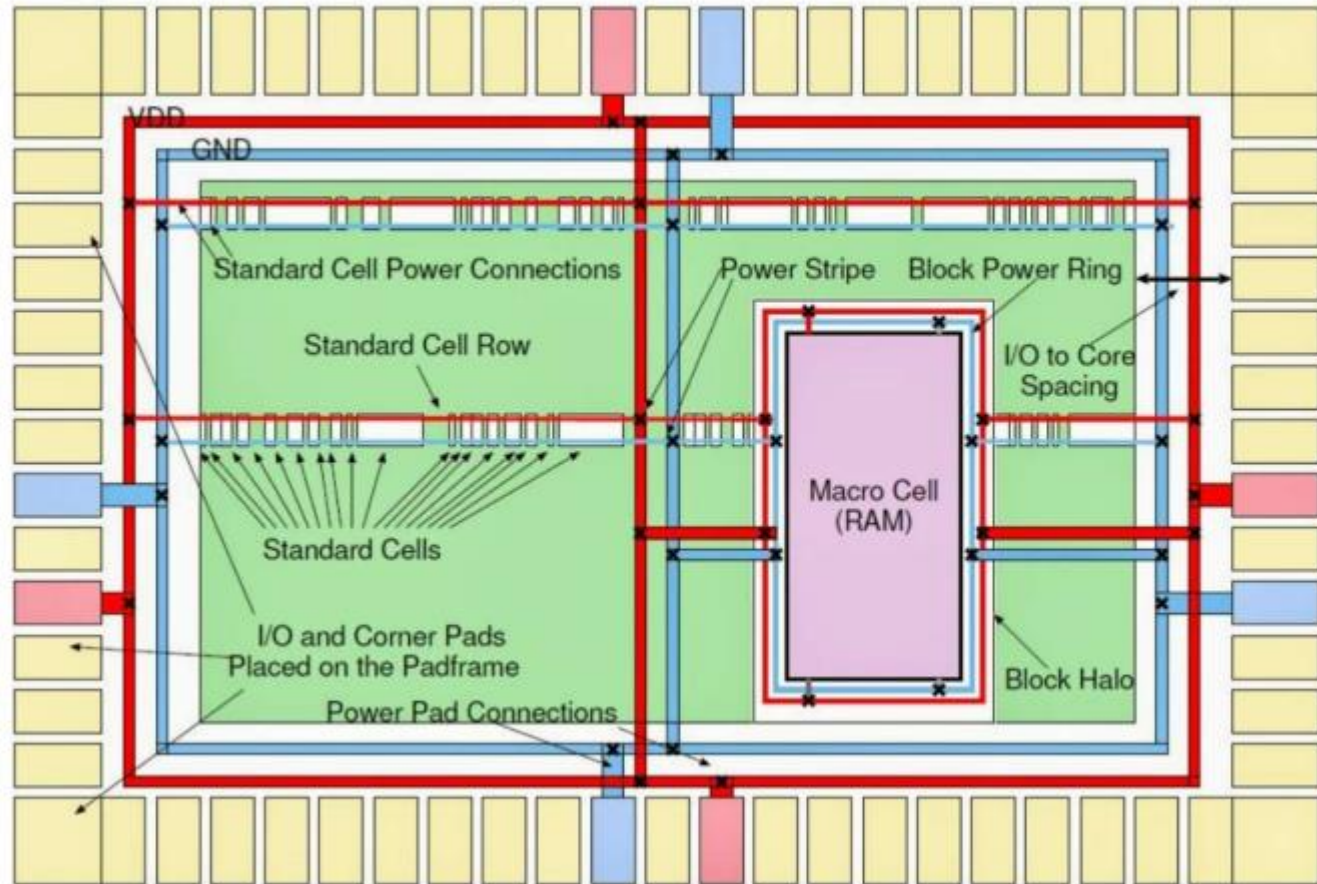
Power plan

Power grid network is created to distribute power to each part of the design efficiently.

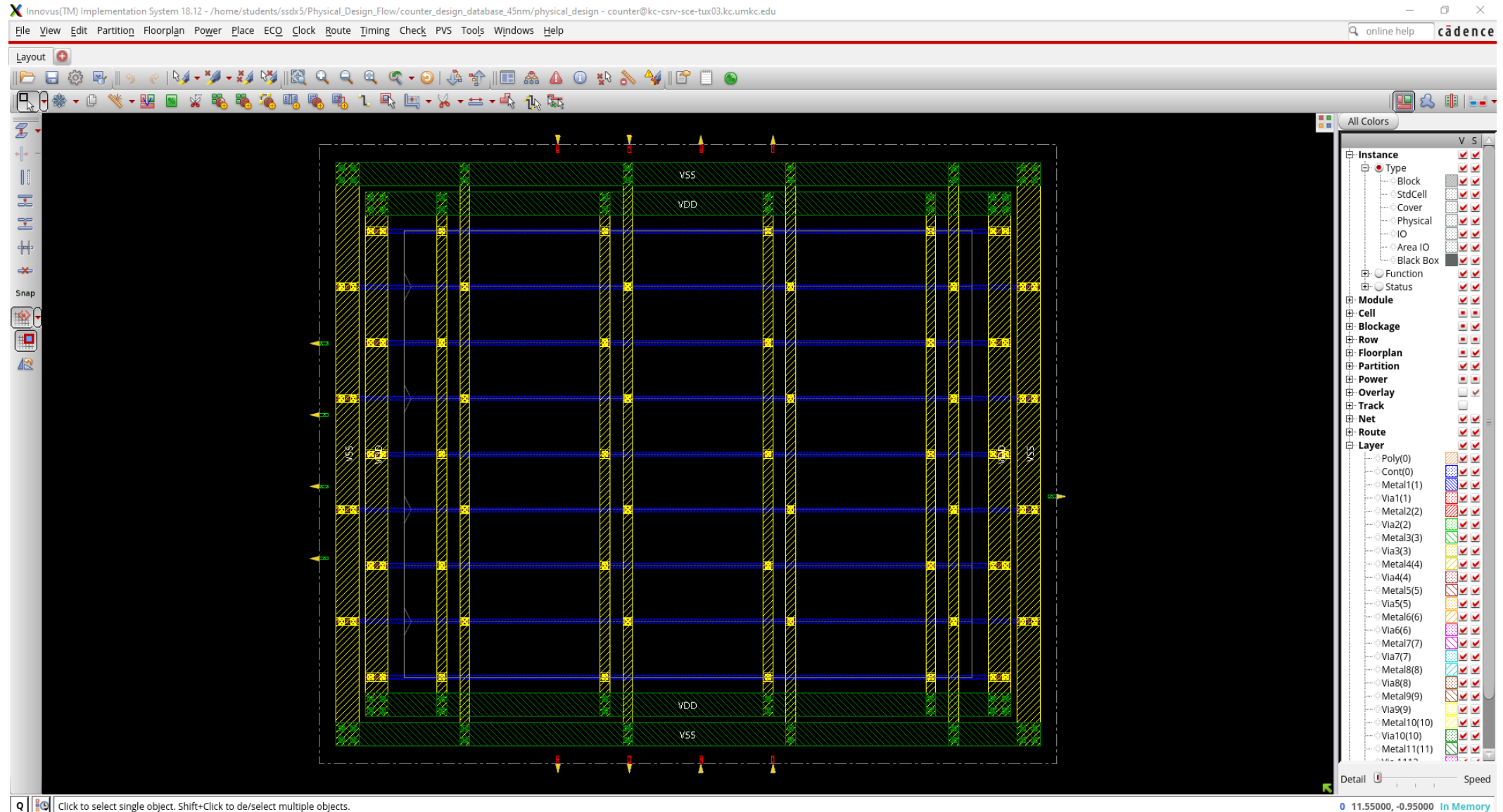
Rings : Carries VDD and VSS around the full-chip

Stripes/Straps : Carries VDD and VSS from Rings across the chip/block

Rails : Connect VDD and VSS to the standard cell VDD and VSS.

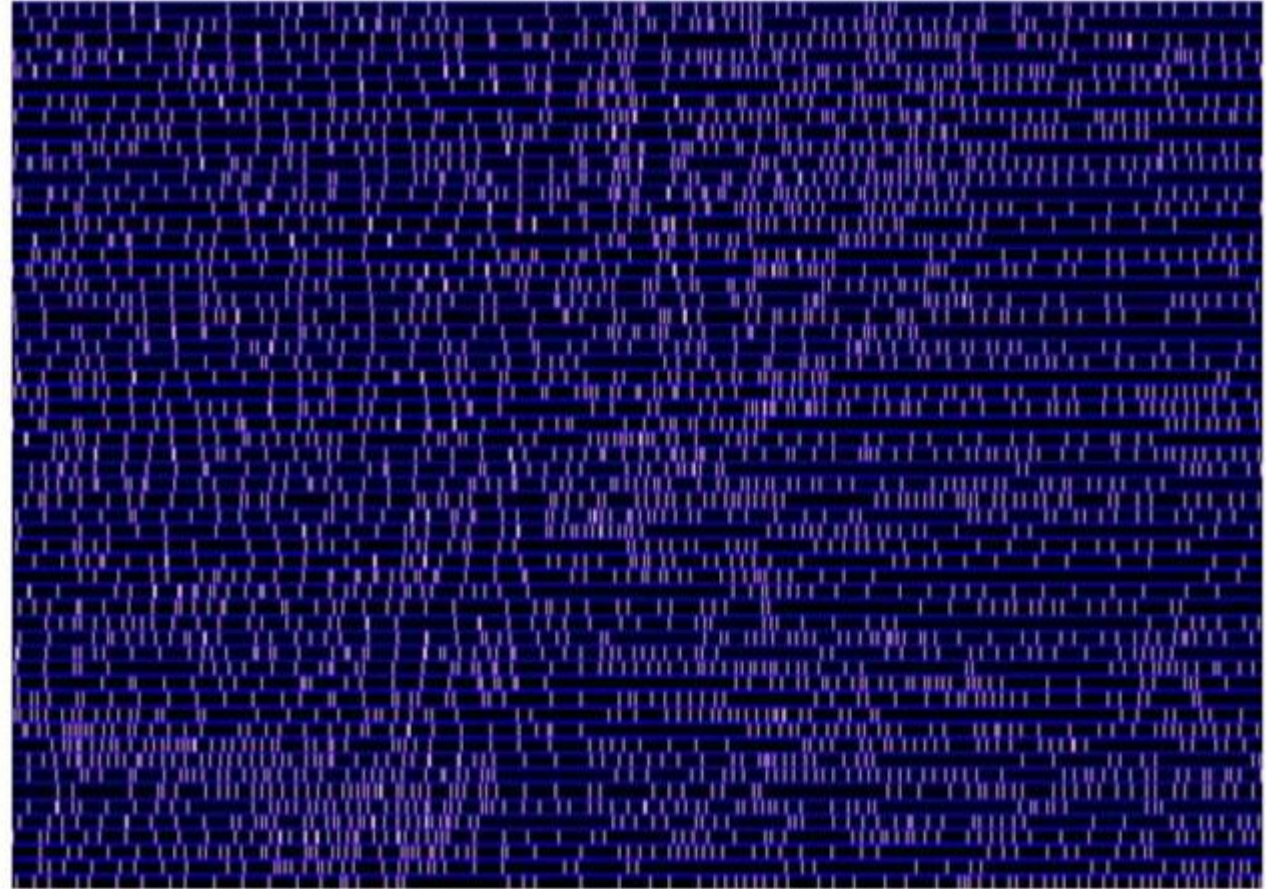


Power plan

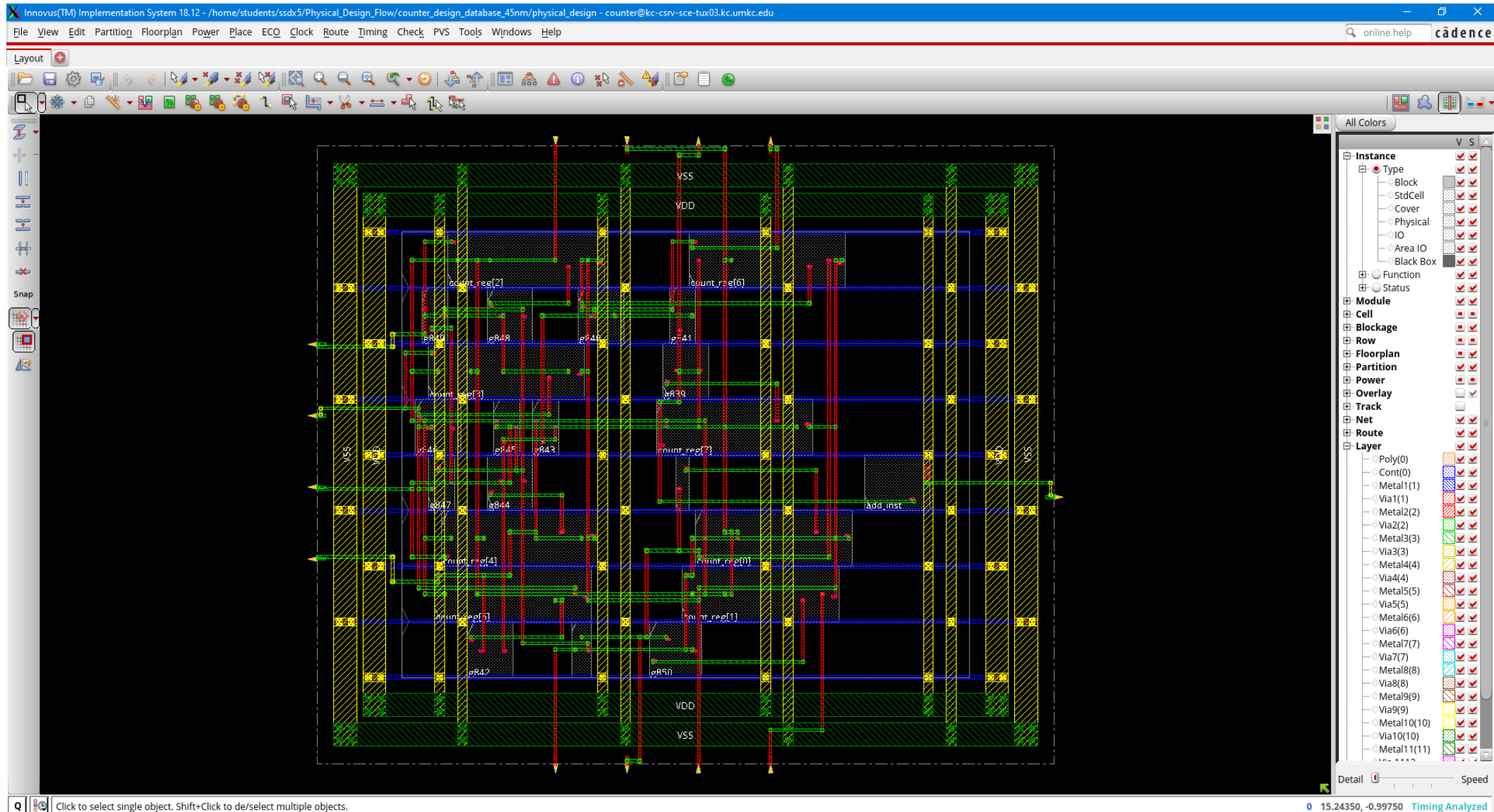


Placement and Optimization

- Placement is the process of placing standard cells in the rows created at powerplan stage.
- Goal is to timing clean & routable design.
- Quality of routing is highly determined by placement.
- Three stages : Global Placement, Legalize Placement and Detailed Placement.
- Design is optimized for Timing, Congestion, Power, Area



Placement and Optimization



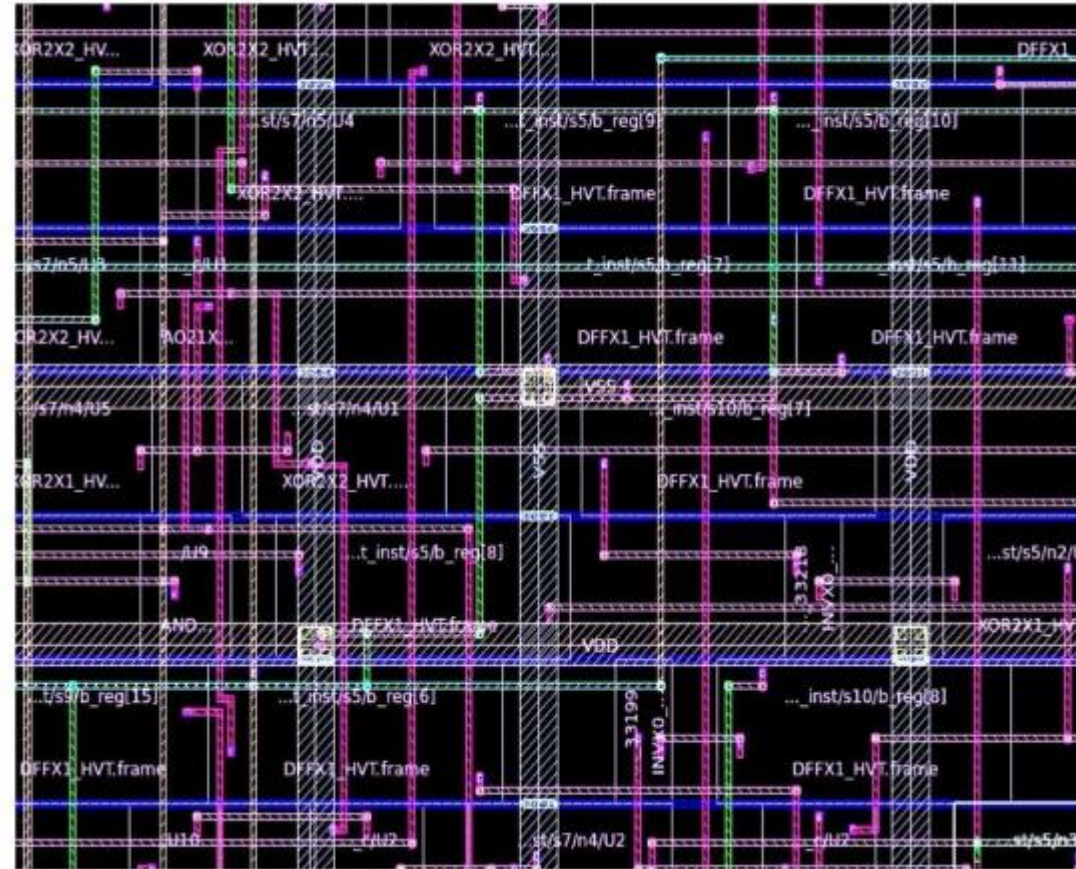
CTS and Optimization

- Clock Tree Synthesis is a process which makes sure that the clock gets distributed evenly to all sequential elements in a design.
- The goal of CTS is to minimize skew and latency.
- The placement data will be given as input for CTS, along with the clock tree constraints.
- Design is optimized for Timing, Congestion & Power

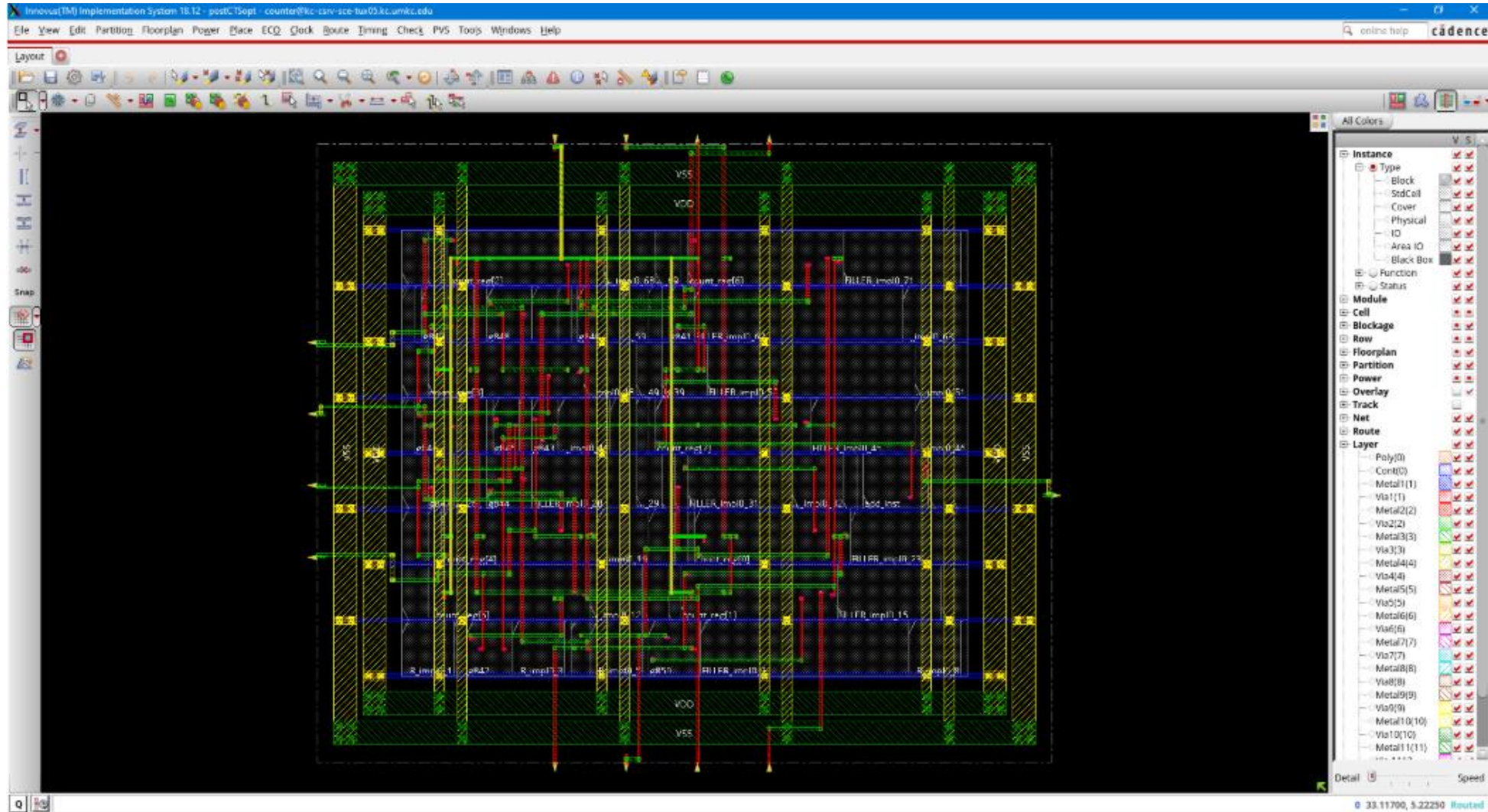


Routing and Optimization

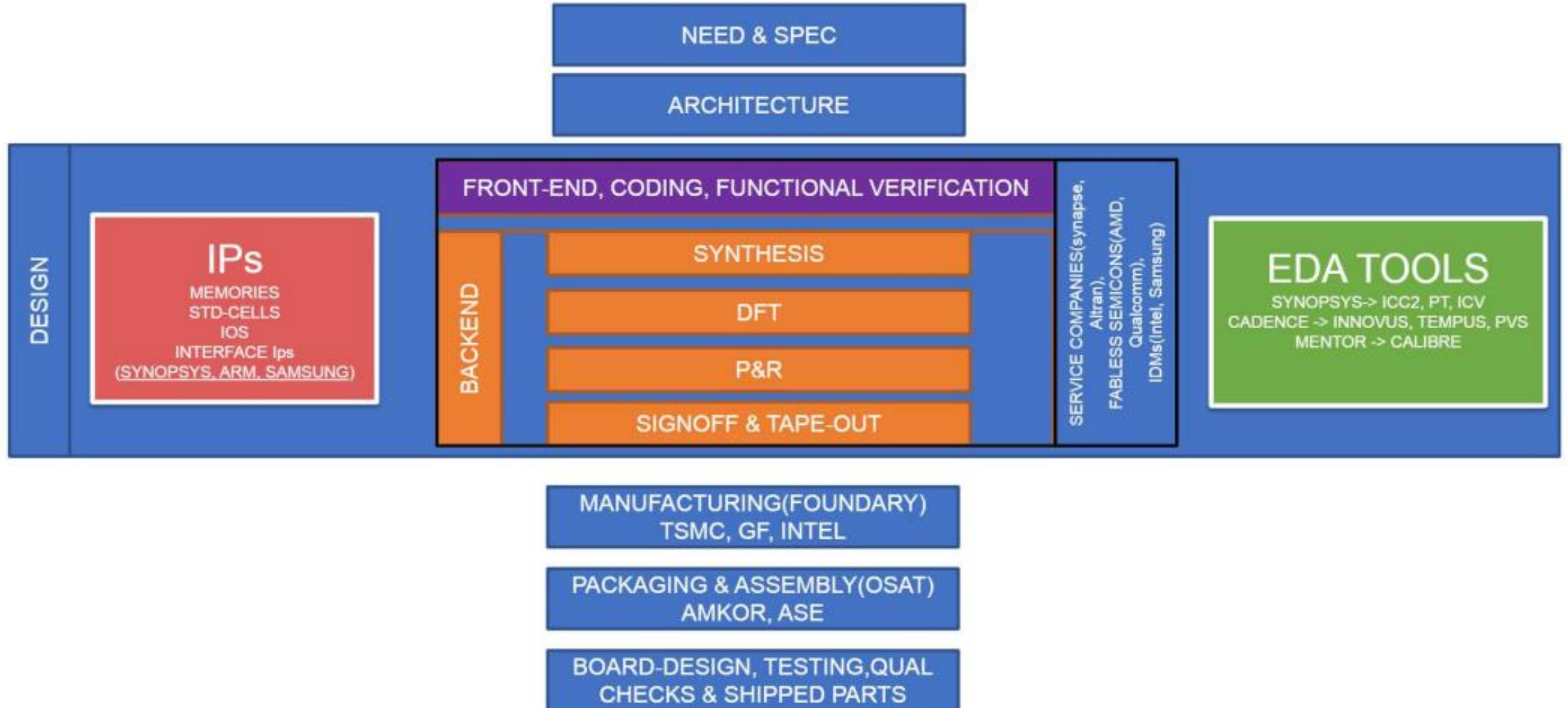
- Involves generating metal wires to connect the pins of same signal while obeying manufacturing design rules.
- There are 3 stages in routing
 - Global Routing
 - Track Assignment
 - Detail Routing
- Design is optimized for timing, power & to be DRC clean



Routing and Optimization



ASIC Design ECOSYSTEM



Questions?

